Exercise 3.2: Analyzing Network Infrastructure with NET Analyzer

Objective
The objective of this experiment is to analyze a local Wi-Fi network using the NET Analyzer mobile application and to understand its configuration, performance, and potential issues.

Tools and Device
Device: Apple iPhone
Application: NET Analyzer (iOS)

Network Information Collected
The device was connected to a local Wi-Fi network. The default gateway and DNS server were identified as 192.168.1.1, indicating a private home router configuration. The assigned IP address of the device was 192.168.1.43 with a subnet mask of 255.255.255.0, confirming a standard Class C private network.

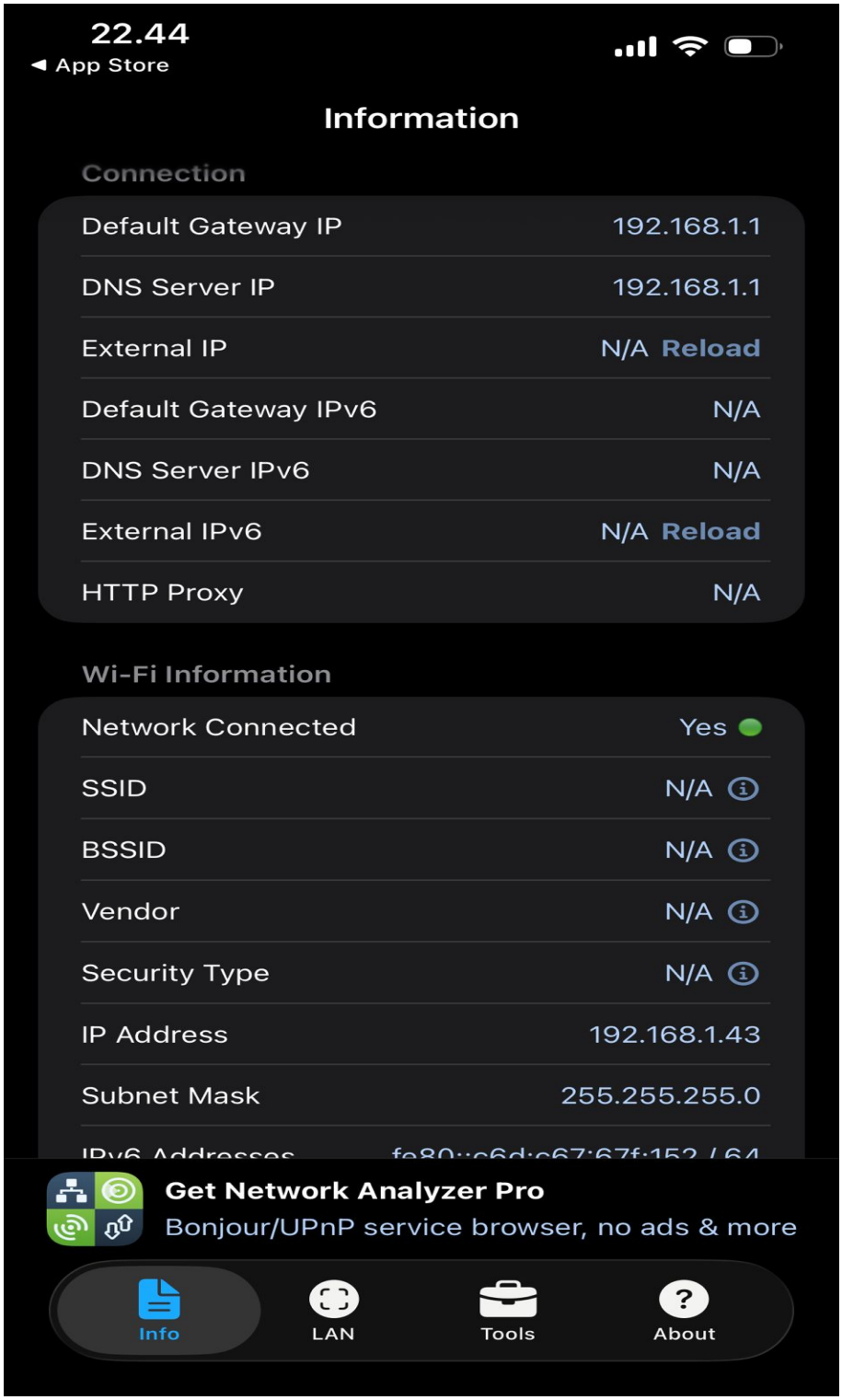
Observations
The Wi-Fi connection was active and stable at the time of measurement. IPv6 connectivity was not available. SSID, BSSID, and security parameters were not displayed by the application due to permission or privacy limitations in iOS.

Analysis and Troubleshooting
The use of a single gateway and DNS server suggests centralized routing and name resolution. The absence of IPv6 indicates reliance on IPv4-only communication. Missing SSID and BSSID information limits deeper interference and channel analysis.

Recommendations
To improve analysis accuracy, full application permissions should be enabled. Channel analysis should be conducted to avoid interference. If congestion is detected, router repositioning or channel changes are recommended.

Conclusion
This experiment demonstrates how NET Analyzer can be used to inspect local Wi-Fi network infrastructure. While basic IP-level parameters were successfully identified, additional permissions are required for complete wireless-layer analysis.

Figure: NET Analyzer – Wi-Fi Network Information


22.44
App Store
Information
Connection
Default Gateway IP 192.168.1.1
DNS Server IP 192.168.1.1
External IP N/A Reload
Default Gateway IPv6 N/A
DNS Server IPv6 N/A
External IPv6 N/A Reload
HTTP Proxy N/A
Wi-Fi Information
Network Connected Yes
SSID N/A
BSSID N/A
Vendor N/A
Security Type N/A
IP Address 192.168.1.43
Subnet Mask 255.255.255.0
Get Network Analyzer Pro
Bonjour/UPnP service browser, no ads & more
Info
LAN
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